QUANTITATIVE RESEARCH MEMORANDUM

Discrete-Time Option Pricing & Asymptotic Convergence

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Executive Summary

This memo evaluates discrete-time option pricing via the Cox-Ross-Rubinstein (CRR) binomial lattice, deriving the synthetic risk-neutral measure directly from core no-arbitrage principles. We contrast the symmetric jump mechanics of the CRR parametrization with the drift-corrected Jarrow-Rudd (JR) approach. Finally, analytical proofs and numerical data confirm that as the time step $\Delta t \to 0$, the discrete CRR valuation asymptotically converges to the continuous Black-Scholes-Merton (BSM) baseline.

PART A: LATTICE MECHANICS & PARAMETRIZATIONS

1.1 Market Setup and Assumptions

Let $(\Omega, \mathscr{F}, P)$ be a filtered probability space supporting the risk-free asset B and the risky asset S. We operate under the standard no-arbitrage framework with the following assumptions: (i) frictionless markets with trading occurring at discrete intervals Δt; (ii) constant risk-free rate $r \geq 0$; (iii) constant volatility $\sigma > 0$; (iv) no dividends; and (v) no transaction costs or taxes.

We partition the option's life $[0,T]$ into N equal time steps of length $\Delta t = T/N$. At each node the asset price can move to one of two states: *up* by factor $u > 1$ or *down* by factor $d \in (0,1)$. The gross return of the risk-free asset over Δt is $R = e^{r\Delta t}$.

1.2 Derivation of the Binomial Tree Framework

1.2.1 Asset Price Dynamics on the Lattice

Starting from an initial asset price S_0, the price at lattice node (n, j), step n with j up-moves, is given by:

$$S(n, j) = S_0 \cdot u^j \cdot d^{n-j}, \qquad j = 0,1,\ldots,n; \quad n = 0,1,\ldots,N. \tag{1}$$

The *recombining* property ($u \cdot d = 1$ in the CRR case) ensures the tree has $(N+1)(N+2)/2$ unique nodes rather than 2^N, making computation tractable.

1.2.2 Risk-Neutral Probability Measure

Instead of relying immediately on the First Fundamental Theorem, we derive the risk-neutral measure by constructing a riskless replicating portfolio. Consider a portfolio Π consisting of Δ shares of the underlying asset and a cash amount B invested at the risk-free rate. Equating the portfolio's value at time $t = \Delta t$ to the option's payoffs in the up (V_u) and down (V_d) states:

$$\Pi_u = \Delta \cdot (uS) + Be^{r\Delta t} = V_u,$$
$$\Pi_d = \Delta \cdot (dS) + Be^{r\Delta t} = V_d.$$

Solving this linear system for the hedge ratio Δ and the cash position B yields:

$$\Delta = \frac{V_u - V_d}{S(u-d)}, \qquad B = \frac{uV_d - dV_u}{e^{r\Delta t}(u-d)}. \tag{2}$$

To prevent arbitrage, the option price today must equal the cost of setting up this replicating portfolio, $V = \Delta S + B$. Substituting Δ and B and rearranging the terms:

$$V = e^{-r\Delta t}\left[\left(\frac{e^{r\Delta t} - d}{u-d}\right)V_u + \left(\frac{u - e^{r\Delta t}}{u-d}\right)V_d\right]. \tag{3}$$

This implies the existence of a synthetic risk-neutral probability measure $\mathbb{Q}$, where the probability of an up-move q emerges naturally as:

$$q \;=\; \frac{e^{r\Delta t} - d}{u-d}. \tag{4}$$

For $\mathbb{Q}$ to be a valid probability measure ($0 < q < 1$), the no-arbitrage condition $d < e^{r\Delta t} < u$ must strictly hold. Under $\mathbb{Q}$, the European option value satisfies:

$$V(n,j) \;=\; e^{-r\Delta t}\big[q \cdot V(n+1,\, j+1) + (1-q)\cdot V(n+1,\, j)\big], \tag{5}$$

with terminal condition $V(N,j) = \Psi\big(S(N,j)\big)$, where Ψ is the option payoff function. For European calls: $\Psi(S) = \max(S-K,0)$; for European puts: $\Psi(S) = \max(K-S,0)$.

1.2.3 American Option Pricing with Early Exercise

American options grant the holder the right to exercise at any lattice node. At each node (n,j), the holder compares the intrinsic immediate exercise value $E(n,j)$ with the continuation value $C(n,j)$, defined as the discounted expected payoff of keeping the option alive for one more step:

$$C(n,j) = e^{-r\Delta t}[q \cdot V^{Am}(n+1,j+1) + (1-q)\cdot V^{Am}(n+1,j)] \tag{6}$$

The American option value at that node is therefore the maximum of the two:

$$V^{Am}(n,j) = \max\{E(n,j), C(n,j)\} \tag{7}$$

where $E(n,j) = \max(S(n,j)-K,0)$ for American calls and $E(n,j) = \max(K - S(n,j),0)$ for American puts. The early-exercise boundary is determined dynamically via backward induction, making binomial trees particularly suited for American option pricing.

1.3 CRR vs. Jarrow-Rudd Parametrization

1.3.1 Cox-Ross-Rubinstein (CRR) — Equal Jump Sizes

CRR (1979) set u and d symmetrically around unity, ensuring the lattice recombines. The jump sizes are constructed strictly to match the variance of the geometric Brownian motion log-return, leaving the drift to be absorbed by the probabilities:

$$u_{\text{CRR}} = e^{\sigma\sqrt{\Delta t}}, \qquad d_{\text{CRR}} = e^{-\sigma\sqrt{\Delta t}} = \frac{1}{u_{\text{CRR}}}. \tag{8}$$

This guarantees the recombination property $u \cdot d = 1$. The risk-neutral probability under CRR is:

$$q_{\text{CRR}} = \frac{e^{r\Delta t} - e^{-\sigma\sqrt{\Delta t}}}{e^{\sigma\sqrt{\Delta t}} - e^{-\sigma\sqrt{\Delta t}}}. \tag{9}$$

To demonstrate the drift correction embedded within q_{CRR}, we apply a Taylor series expansion to the exponential terms for small Δt. Expanding $e^x \approx 1 + x + \frac{x^2}{2}$ yields:

$$u_{\text{CRR}} \approx 1 + \sigma\sqrt{\Delta t} + \frac{\sigma^2 \Delta t}{2}, \tag{10}$$

$$d_{\text{CRR}} \approx 1 - \sigma\sqrt{\Delta t} + \frac{\sigma^2 \Delta t}{2}, \tag{11}$$

$$e^{r\Delta t} \approx 1 + r\Delta t. \tag{12}$$

Substituting these approximations into the definition of q_{CRR} in (9):

$$q_{\text{CRR}} \approx \frac{(1 + r\Delta t) - \left(1 - \sigma\sqrt{\Delta t} + \frac{1}{2}\sigma^2 \Delta t\right)}{\left(1 + \sigma\sqrt{\Delta t} + \frac{1}{2}\sigma^2 \Delta t\right) - \left(1 - \sigma\sqrt{\Delta t} + \frac{1}{2}\sigma^2 \Delta t\right)} = \frac{\sigma\sqrt{\Delta t} + \left(r - \frac{1}{2}\sigma^2\right)\Delta t}{2\sigma\sqrt{\Delta t}}. \tag{13}$$

Simplifying this fraction provides the second-order approximation:

$$q_{\text{CRR}} \approx \frac{1}{2} + \frac{(r - \sigma^2/2)\sqrt{\Delta t}}{2\sigma}. \tag{14}$$

This explicitly demonstrates how the risk-neutral drift $(r - \sigma^2/2)$ is absorbed directly into the probability measure, rather than the jump sizes, under the CRR parametrization.

1.3.2 Jarrow-Rudd (JR) — Equal Risk-Neutral Probabilities

Jarrow and Rudd (1983) force the risk-neutral probabilities to be equal by setting $q = \frac{1}{2}$ by construction. To ensure the lattice avoids arbitrage and converges to the geometric Brownian motion log-return distribution, the jump sizes must absorb the Itô-corrected drift. We establish this by matching the mean and variance of the one-step log-returns under the risk-neutral measure $\mathbb{Q}$:

$$\mathbb{E}_{\mathbb{Q}}\left[\ln \frac{S_{\Delta t}}{S_0}\right] = \frac{1}{2}\ln u_{\text{JR}} + \frac{1}{2}\ln d_{\text{JR}} = \left(r - \frac{\sigma^2}{2}\right)\Delta t, \tag{15}$$

$$\text{Var}_{\mathbb{Q}}\left[\ln \frac{S_{\Delta t}}{S_0}\right] = \frac{1}{2}(\ln u_{\text{JR}})^2 + \frac{1}{2}(\ln d_{\text{JR}})^2 - \left(\left(r - \frac{\sigma^2}{2}\right)\Delta t\right)^2 = \sigma^2 \Delta t. \tag{16}$$

Solving this system of equations for the jump sizes directly yields:

$$u_{\text{JR}} = e^{(r-\sigma^2/2)\Delta t + \sigma\sqrt{\Delta t}}, \qquad d_{\text{JR}} = e^{(r-\sigma^2/2)\Delta t - \sigma\sqrt{\Delta t}}. \tag{17}$$

Note $u_{\text{JR}} \cdot d_{\text{JR}} = e^{2(r-\sigma^2/2)\Delta t} \neq 1$ in general, so the JR tree is not symmetric around the initial spot price. However, the Itô-corrected drift $(r - \sigma^2/2)$ is now explicitly incorporated into the jump sizes rather than the probabilities.

1.3.3 Comparative Analysis

Table 1: Comparative summary of CRR and Jarrow-Rudd parametrizations.

Property	CRR (1979)	Jarrow-Rudd (1983)
Jump up	$u = e^{\sigma\sqrt{\Delta t}}$	$u = e^{(r-\sigma^2/2)\Delta t+\sigma\sqrt{\Delta t}}$
Jump down	$d = e^{-\sigma\sqrt{\Delta t}}$	$d = e^{(r-\sigma^2/2)\Delta t-\sigma\sqrt{\Delta t}}$
Risk-neutral prob.	$q = (e^{r\Delta t}-d)/(u-d)$	$q = 1/2$ (by construction)
Recombination	Yes ($ud = du$)	Yes ($ud = du$)
Symmetry	Symmetric ($ud = 1$)	Asymmetric ($ud \neq 1$)
Drift correction	Implicit (via q)	Explicit (in u, d)
Convergence rate	$O(\Delta t)$ — first order	$O(\Delta t)$ — first order
Implementation	Simpler (symmetric)	Slightly more complex

Both parametrizations converge to the BSM price as $N \to \infty$ ($\Delta t \to 0$), but exhibit different finite-sample behaviors. CRR prices natively oscillate around the continuous baseline due to discrete strike misalignment. Conversely, as documented by Leisen and Reimer (1996) and empirically verified in Section 2.2, the drift-corrected nature of the Jarrow-Rudd lattice explicitly dampens these oscillations, yielding smoother convergence with lower-amplitude oscillations and consistently lower absolute errors for European options at equivalent finite steps.

Part B: Continuous-Time Convergence

2.1 The Black-Scholes-Merton Model as the Continuous-Time Limit

We now analytically demonstrate that the CRR binomial price converges to the BSM price as $N \to \infty$ (equivalently $\Delta t = T/N \to 0$). The proof proceeds via moment matching and the Central Limit Theorem (CLT), verified analytically via the Binomial Theorem.

2.1.1 Black-Scholes-Merton Closed-Form Price

For a European call option with strike K, maturity T, spot S_0, risk-free rate r, and volatility σ, the BSM price is:

$$C_{\text{BSM}} = S_0 \mathcal{N}(d_1) - Ke^{-rT}\mathcal{N}(d_2), \tag{18}$$

where $\mathcal{N}(\cdot)$ denotes the standard normal CDF and:

$$d_1 = \frac{\ln(S_0/K) + (r+\sigma^2/2)\,T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}. \tag{19}$$

2.1.2 CRR Price as a Binomial Sum

The CRR price for a European call, obtained by backward induction and application of the Binomial Theorem, admits the closed-form representation:

$$C_{\text{CRR}}(N) = S_0\,\Phi(a; N, q') - Ke^{-rT}\,\Phi(a; N, q), \tag{20}$$

where

$$\Phi(a; N, q) = \sum_{j=a}^{N} \binom{N}{j} q^j (1-q)^{N-j}$$

is the complementary binomial CDF, $a = \min\{j \in \mathbb{Z} : S_0 u^j d^{N-j} > K\}$ is the minimum number of up-moves for the option to expire in-the-money. The term $q' = q \cdot u \cdot e^{-r\Delta t}$ is the stock-measure probability, arising mathematically from a change of numeraire from the money market account to the underlying stock itself.

2.1.3 Asymptotic Convergence Proof

Theorem 1 (CRR Convergence). *As $N \to \infty$ with $\Delta t = T/N \to 0$:*

$$\left|C_{\text{CRR}}(N) - C_{\text{BSM}}\right| \;=\; O\Big(N^{-1/2}\Big). \tag{21}$$

Proof. **Step 1 — Moment matching.** Under the CRR parametrization, the log-return over one step equals $\ln u = \sigma\sqrt{\Delta t}$ with probability q and $\ln d = -\sigma\sqrt{\Delta t}$ with probability $1-q$ under $\mathbb{Q}$. The mean and variance of the one-step log-return are:

$$\mathbb{E}_{\mathbb{Q}}\left[\ln \tfrac{S_{\Delta t}}{S_0}\right] = (2q-1)\,\sigma\sqrt{\Delta t} = \left(r - \tfrac{\sigma^2}{2}\right)\Delta t + O\Big(\Delta t^{3/2}\Big), \tag{22}$$

$$\operatorname{Var}_{\mathbb{Q}}\left[\ln \tfrac{S_{\Delta t}}{S_0}\right] = 4q(1-q)\,\sigma^2\Delta t = \sigma^2\Delta t + O\big(\Delta t^2\big). \tag{23}$$

Hence CRR matches the GBM log-return mean $(r-\sigma^2/2)\Delta t$ and variance $\sigma^2\Delta t$ to leading order.

Step 2 — CLT application. The N-step log-return is the sum of N i.i.d. scaled Bernoulli random variables taking values $\sigma\sqrt{\Delta t}$ and $-\sigma\sqrt{\Delta t}$. By the CLT, as $N \to \infty$, where the substitution $N\Delta t = T$ yields the stated continuous mean and variance:

$$\ln \frac{S_T}{S_0} = \sum_{k=1}^{N} X_k \xrightarrow{\mathscr{D}} \mathscr{N}\left(\left(r - \tfrac{\sigma^2}{2}\right)T,\; \sigma^2 T\right). \tag{24}$$

Step 3 — Convergence of binomial CDFs. The complementary binomial distribution $\Phi(a; N, q)$ in (20) converges to the normal CDF by the CLT. Standardising the binomial sum yields:

$$z = \frac{j - Nq}{\sqrt{Nq(1-q)}} \xrightarrow{\mathscr{D}} \mathscr{N}(0,1). \tag{25}$$

The critical threshold a represents the minimum number of up-moves required for the option to expire in-the-money, meaning $S_0 u^a d^{N-a} = K$. Substituting the CRR parameters $u = e^{\sigma\sqrt{\Delta t}}$ and $d = e^{-\sigma\sqrt{\Delta t}}$ and solving for a:

$$a = \frac{\ln(K/S_0) - N\ln d}{\ln(u/d)} = \frac{\ln(K/S_0) + N\sigma\sqrt{\Delta t}}{2\sigma\sqrt{\Delta t}}. \tag{26}$$

We standardise this threshold a using the mean Nq and variance $Nq(1-q)$. Applying the Taylor approximation for q_{CRR} derived in (14) and substituting $\Delta t = T/N$:

$$z_a = \frac{a - Nq}{\sqrt{Nq(1-q)}} \approx \frac{\left(\frac{\ln(K/S_0)}{2\sigma\sqrt{\Delta t}} + \frac{N}{2}\right) - \left(\frac{N}{2} + \frac{(r-\sigma^2/2)N\Delta t}{2\sigma\sqrt{\Delta t}}\right)}{\frac{1}{2}\sqrt{N}}. \tag{27}$$

Notice that the $\frac{N}{2}$ terms cancel. Substituting $N\Delta t = T$, simplifying the denominator, and taking the continuous limit as $N \to \infty$ ($\Delta t \to 0$), the discrete threshold z_a converges exactly to the continuous BSM boundary:

$$z_a \longrightarrow \frac{\ln(K/S_0) - (r - \sigma^2/2)T}{\sigma\sqrt{T}} = -d_2. \tag{28}$$

Because $\Phi(a;N,q)$ represents the probability of exceeding this threshold, we apply the right-tail symmetry of the standard normal distribution:

$$\Phi(a;N,q) = P(Z \geq z_a) \longrightarrow P(Z \geq -d_2) = \mathcal{N}(d_2). \tag{29}$$

A parallel derivation for the stock-measure probability $q' = q \cdot u \cdot e^{-r\Delta t}$ reveals that the Taylor expansion shifts the probability by an additional $\sigma\sqrt{\Delta t}$, yielding a crucial sign flip on the variance term:

$$q' \approx \frac{1}{2} + \frac{(r + \sigma^2/2)\sqrt{\Delta t}}{2\sigma}. \tag{30}$$

This explicit sign flip alters the standardisation mean to exactly recover $-d_1$. Therefore:

$$\Phi(a;N,q) \;\to\; \mathcal{N}(d_2), \qquad \Phi(a;N,q') \;\to\; \mathcal{N}(d_1),$$

so $C_{\text{CRR}}(N) \to C_{\text{BSM}}$ as $N \to \infty$.

Convergence rate. The error $|C_{\text{CRR}}(N) - C_{\text{BSM}}| = O(1/\sqrt{N})$ follows from the Berry-Esseen theorem applied to the binomial approximation. Leisen and Reimer (1996) showed that for odd N centred at the strike, the rate improves to $O(1/N)$. □

2.2 Numerical Illustration of Convergence

While the asymptotic proof in Section 2.1 is derived for European calls, convergence of the put price follows immediately via put-call parity, $C - P = S_0 - Ke^{-rT}$. To illustrate this empirically, consider an ATM European put with $S_0 = 100$, $K = 100$, $T = 1$ yr, $r = 5\%$, $\sigma = 20\%$. The exact continuous BSM put baseline is $P_{\text{BSM}} = 5.5735$.

Table 2: CRR vs. JR convergence for the ATM European put. The JR parametrization consistently yields lower absolute errors at equivalent step counts compared to CRR.

N (Steps)	CRR Price	CRR Error	JR Price	JR Error
5	5.9289	0.3554	5.8813	0.3078
10	5.3764	0.1972	5.6491	0.0756
25	5.6439	0.0704	5.5652	0.0083
50	5.5336	0.0399	5.6107	0.0371
100	5.5536	0.0200	5.5830	0.0095
200	5.5635	0.0100	5.5683	0.0052
500	5.5695	0.0040	5.5763	0.0028
1000	5.5715	0.0020	5.5751	0.0016
∞ (BSM)	**5.5735**	—	**5.5735**	—

As shown in Table 2, the raw CRR prices exhibit actual oscillatory convergence around the BSM baseline. While the Berry-Esseen theorem provides a universal global bound of $O(1/\sqrt{N})$, the geometric alignment of CRR terminal nodes precisely on the strike for At-The-Money options actively neutralizes first-order discrete distribution errors. This strike-centering explicitly drives the accelerated local convergence rate of $O(1/N)$ observed in our empirical data.

2.3 Convergence of American Option Prices

The convergence result for European options extends to American options via optimal stopping theory. The American option price satisfies:

$$V^{Am} = \sup_{\tau \in [0,T]} \mathbb{E}_{\mathbb{Q}}\left[e^{-r\tau}\Psi(S_\tau)\right]. \tag{31}$$

Unlike European options, American options generally lack a universal closed-form continuous solution. The necessity of the lattice approach stems directly from the free-boundary nature of this pricing problem. In the continuous limit, the American put price satisfies the Black-Scholes PDE strictly within the continuation region $S > S^*(t)$, where $S^*(t)$ is the critical early-exercise boundary. At this boundary, the option must satisfy both the value-matching condition ($V = K - S$) and the smooth-pasting condition ($\partial V/\partial S = -1$). Because $S^*(t)$ is unknown *a priori* and must be solved simultaneously with the option price, the analytical pricing fails. The binomial lattice elegantly circumvents this by embedding the optimal stopping decision at every discrete node via backward induction, allowing the dynamic free boundary to emerge implicitly. Conceptually, the total value of the American put can be decomposed into the European put value plus the Early Exercise Premium (EEP):

$$V^{Am}(N) = V^{Eu}(N) + \text{EEP}(N) \tag{32}$$

The backward induction algorithm captures this EEP(N) directly via the $\max\{\cdot\}$ operator function applied during backward induction as defined in equation (7).

Theorem 2 (American Convergence). *Let $V^{\text{Am}}_{\text{CRR}}(N)$ denote the CRR price of an American option with N steps. Then:*

$$V^{Am}_{\text{CRR}}(N) \longrightarrow V^{Am} \quad \text{as } N \to \infty,$$

at rate $O(1/\sqrt{N})$.

As the number of time steps increases, the discrete exercise opportunities become dense in time, allowing the binomial stopping problem to naturally converge to the continuous boundary of the American option (Amin and Khanna, 1994).

Table 3: Convergence of American vs. European Put prices. The high-density $N = 5000$ CRR price is used as the true baseline reference due to the lack of a closed-form continuous solution for early exercise.

N (Steps)	CRR European	CRR American	JR European	JR American
5	5.9289	6.3678	5.8813	6.3475
10	5.3764	6.0043	5.6491	6.1103
25	5.6439	6.1462	5.5652	6.1014
50	5.5336	6.0737	5.6107	6.1181
100	5.5536	6.0824	5.5830	6.1000
200	5.5635	6.0864	5.5683	6.0873
500	5.5695	6.0888	5.5763	6.0928
1000	5.5715	6.0896	5.5751	6.0916
Ref. ($N = 5000$)	—	**6.0902**	—	—

2.4 Key Theoretical Properties and Implications

- **No-arbitrage consistency.** Both CRR and JR models are arbitrage-free by construction ($d < e^{r\Delta t} < u$), and the risk-neutral measure $\mathbb{Q}$ is unique in the binomial setting, guaranteeing a unique pricing functional.

- **Oscillatory convergence.** For fixed strike and maturity, $C_{\text{CRR}}(N)$ does not converge monotonically; it oscillates around C_{BSM}. This occurs because the terminal nodes of the discrete lattice rarely fall exactly on the strike price K. The non-linearity of the payoff function at K causes the discrete pricing engine to systematically over- or under-estimate the continuous price depending on the grid's alignment with K at step N. The smoothed sequence $\frac{1}{2}[C_{\text{CRR}}(N) + C_{\text{CRR}}(N+1)]$ averages these discrete strike misalignments and converges monotonically.

- **Richardson extrapolation.** Using two lattice prices $V(N)$ and $V(2N)$, the estimate $2V(2N) - V(N)$ has theoretical error $O(1/N)$ rather than $O(1/\sqrt{N})$, substantially improving accuracy at low computational cost.

Together, these theoretical properties and empirical validations confirm the CRR lattice as a robust, auditable pricing engine suitable for production deployment across both European and American equity derivatives.

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